Math 571, Sample Midterm Exam 1, February 25, 2025

Instructions

1. This exam consists of this cover sheet, then pages numbered 2 to 8. It has 6 multi-part questions. Ensure that your exam has all pages.
2. The only aid permitted is one "note card" on one side of US Letter or A4 paper and a calculator approved for use on the SAT.
3. We recommend that you read through the exam before beginning. **Note that some questions are worth more than other questions.**
4. Answer the questions in the boxes provided on the question sheets. If you run out of room for an answer, spilling over the edge of the box slightly is preferable to writing elsewhere, but in any case we'll read what you write, eventually.
5. Simplify expressions (or leave unsimplified) in a way that clearly expresses steps of computation and intuitive concepts, even if that is not the "official" way to simplify. For example, $\frac{1}{\sqrt{14}}\begin{bmatrix}1\\2\\3\end{bmatrix} = \begin{bmatrix}\frac{\sqrt{14}}{14}\\ \frac{\sqrt{14}}{7}\\ \frac{3\sqrt{14}}{14}\end{bmatrix}$, but the expression on the left more cleanly conveys that the vector is propotional to $\begin{bmatrix}1\\2\\3\end{bmatrix}$ and has norm 1.
6. Have fun!

Name: ____________________

UMID: ____________________

Uniqname: ____________________

Question:	1	2	3	4	5	6	Total
Points:	5	1	1	1	1	4	13
Bonus Points:	0	0	0	0	0	0	0
Score:							

Todo: Make answer boxes or spaces, cut problems for time, prettify numbers

Question 1 (5 points)

Suppose $A = \begin{bmatrix} 1 & -1 \\ 1 & 5 \\ 0 & 6 \end{bmatrix}$

In the following, explain key steps, commenting on runtime. Illustrate the steps in a way that generalizes to bigger matrices and higher dimensions.

(a) (1 point) Find a reduced QR factorization of A by classical Gram-Schmidt (which orthonomalizes columns 1 to j of A before considering column $j+1$). Make the diagonal elements of R non-negative reals.

Solution: Classical Gram-Schmidt has us first normalize the left column of A, project the second column **off** the span of the previous columns, then normalize. These actions can be recorded in upper triangular matrices, whose product and inverses are also upper triangular. We have

$$\begin{bmatrix} 1 & -1 \\ 1 & 5 \\ 0 & 6 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2/\sqrt{2} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1/\sqrt{54} \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{6} \\ 1/\sqrt{2} & 1/\sqrt{6} \\ 0 & 2/\sqrt{6} \end{bmatrix}.$$

The inverses of the R matrices are of the same form—shear in the opposite direction, scale by the reciprocal. So

$$\begin{aligned} \begin{bmatrix} 1 & -1 \\ 1 & 5 \\ 0 & 6 \end{bmatrix} &= \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{6} \\ 1/\sqrt{2} & 1/\sqrt{6} \\ 0 & 2/\sqrt{6} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & \sqrt{54} \end{bmatrix} \begin{bmatrix} 1 & +2/\sqrt{2} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{2} & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{6} \\ 1/\sqrt{2} & 1/\sqrt{6} \\ 0 & 2/\sqrt{6} \end{bmatrix} \begin{bmatrix} \sqrt{2} & 2\sqrt{2} \\ 0 & \sqrt{54} \end{bmatrix} = QR. \end{aligned}$$

As a check, the columns of Q are orthogonal, ignoring uniform square root denominators. They have norm 1, since $1^2 + 1^2 = 2$ and $1^2 + (-1)^2 + 2^2 = 6$. Successive columns span the same spaces as A since the first columns are proportional and $\begin{bmatrix} -1 \\ 5 \\ 6 \end{bmatrix} - 2\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -3 \\ 3 \\ 6 \end{bmatrix}$, a multiple of the second column of Q. (There are alternative ways to handle the square roots to keep numbers prettier, but that would require additional hacks—though not hard—to express classical Gram-Schmidt, as required.)

(b) (1 point) Let $b = \begin{bmatrix} 2 \\ 8 \\ 15 \end{bmatrix}$. Find x to minimize $\|Ax - b\|$, using a reduced QR factorization. Comment on the runtime (not including time to find the QR factorization), including how that generalizes to bigger matrices.

Solution: We want to solve $QRx = b$. Since Q and A have the same range, we can project onto that range with $QQ^*b = \frac{1}{6}\begin{bmatrix} 4 & 2 & -2 \\ 2 & 4 & 2 \\ -2 & 2 & 4 \end{bmatrix} b = \begin{bmatrix} -1 \\ 11 \\ 12 \end{bmatrix}$, noting that QQ^* is the symmetric 3-by-3 matrix (so, less than full rank, suitable to project 3-vectors onto a plane), gotten by taking all possible dot products of 2-columns of Q^* (rows of Q). We can see that the result is in the span of A, noting that $2 \cdot 6 = 12$, so we need twice the second column, and so that is added to the first column. So $x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

Going from Q to Q^* takes mn time (to copy) or, arguably, Q^* is formed from Q with the QR-factorization, before b is presented, or just multiply Q^*b from Q and b without forming Q^* at all. Multiplying Q^*b takes $O(mn)$ time to look at each entry of Q^*. Multiplying Q by the resulting vector Q^*b also takes $O(mn)$ time. Overall, that is quadratic time, less than the cubic time needed to find the QR-factorization by any of our methods.

(c) (1 point) Find a column, b_3, orthogonal to the range of A. Let $B = [A \quad b_3]$. (No cross product without a **lot** of explanation on how to generalize the cross product to higher dimensions.)

Solution: For b_3 to be orthogonal to $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, we need (arbitrary length) $b_3 = \begin{bmatrix} 1 \\ -1 \\ * \end{bmatrix}$. To make b_3 orthogonal to the other column, we get $b_3 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$. More generally, pick a random vector of ± 1's and, if necessary, $\pm 2, \pm 3, \ldots$, maybe with some zeros, which is unlikely to be in the range of A. Then continue the Gram-Schmidt process.

(d) (1 point) Find a full QR factorization QR of A. Explain.

Solution: Much of the work was done in the previous. We just need to normalize b_3 with factor $\frac{1}{\sqrt{3}}$. We get

$$\begin{bmatrix} 1 & -1 \\ 1 & 5 \\ 0 & 6 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \\ 1/\sqrt{2} & 1/\sqrt{6} & -1/\sqrt{3} \\ 0 & 2/\sqrt{6} & 1/\sqrt{3} \end{bmatrix} \begin{bmatrix} \sqrt{2} & 2\sqrt{2} \\ 0 & \sqrt{54} \\ 0 & 0 \end{bmatrix}.$$

(e) (1 point) Solve $Bx = b$, for the b above.

Solution: The residual above is $\begin{bmatrix} 2 \\ 8 \\ 15 \end{bmatrix} - \begin{bmatrix} -1 \\ 11 \\ 12 \end{bmatrix} = \begin{bmatrix} 3 \\ -3 \\ 3 \end{bmatrix} = 3b_3$. We get $\begin{bmatrix} 1 & -1 & 1 \\ 1 & 5 & -1 \\ 0 & 6 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 8 \\ 15 \end{bmatrix}$.

Question 2 (1 points)

Let $C = \begin{bmatrix} 1 & 5 \\ 1 & 5 \\ 1 & -1 \\ 1 & -1 \end{bmatrix}$. Find a reduced QR factorization of C by Householder (which triangularizes one column at a time of A, choosing a sign or complex unit strategically). Make the diagonal elements of R non-negative reals.

Solution: The left column of C has norm 2. We want to reflect it to $\begin{bmatrix} -2 \\ 0 \\ 0 \\ 0 \end{bmatrix}$, with negative sign opposite the upper left component of C. The displacement vector v is $v = \begin{bmatrix} -2 \\ 0 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = -\begin{bmatrix} 3 \\ 1 \\ 1 \\ 1 \end{bmatrix}$.

A unit vector in that direction is $u = \frac{-1}{\sqrt{12}} \begin{bmatrix} 3 \\ 1 \\ 1 \\ 1 \end{bmatrix}$, and the reflection we want is $Q_1 = I - 2uu^*$.

If we apply this to the left column of C, we get $\begin{bmatrix} -2 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ by construction (we could also check). If we apply Q_1 to the right column c_2, it's more efficient to do $c_2 - 2u(u^*c_2)$ rather than construct the 4-by-4 reflection matrix. And collect two factors of $-\sqrt{12}$. So $v^*c_2 = -\begin{bmatrix} 3 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 5 \\ -1 \\ -1 \end{bmatrix} = -18$ and $c_2 - 2u(u^*c_2) = c_2 - \frac{2}{\|v\|^2} v(v^*c_2) = \begin{bmatrix} 5 \\ 5 \\ -1 \\ -1 \end{bmatrix} - 2\frac{18}{12} \begin{bmatrix} 3 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \\ -1 \\ -1 \end{bmatrix} - \begin{bmatrix} 9 \\ 3 \\ 3 \\ 3 \end{bmatrix} = \begin{bmatrix} -4 \\ 2 \\ -4 \\ -4 \end{bmatrix}$, so $Q_1C = (I - 2uu^*)C = \frac{1}{6} \begin{bmatrix} -3 & -3 & -3 & -3 \\ -3 & 5 & -1 & -1 \\ -3 & -1 & 5 & -1 \\ -3 & -1 & -1 & 5 \end{bmatrix} C = \begin{bmatrix} -2 & -4 \\ 0 & 2 \\ 0 & -4 \\ 0 & 4 \end{bmatrix}$.

Next, work on the southeast submatrix $\begin{bmatrix} 2 \\ -4 \\ -4 \end{bmatrix}$. The norm of that vector is 6 and the topmost sign is positive, so we want to reflect it to $\begin{bmatrix} -6 \\ 0 \\ 0 \end{bmatrix}$. The displacement is $v = \begin{bmatrix} -6 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 2 \\ -4 \\ -4 \end{bmatrix} = \begin{bmatrix} -8 \\ 4 \\ 4 \end{bmatrix}$, whose norm is $4\sqrt{6}$. So the 3-by-3 reflector is $F = (I - \frac{2}{4^2 \cdot 6} vv')$ and the 4-by-4 is $Q_2 = \begin{bmatrix} 1 & 0 \\ 0 & F \end{bmatrix}$. That's $Q_2 = \frac{1}{3} \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & -1 & 2 & 2 \\ 0 & 2 & 2 & -1 \\ 0 & 2 & -1 & 2 \end{bmatrix}$.

UPDATE: Fixed negative sign of R. Inserted matrix inverse and switched the order in $(Q_2Q_1)^{-1}$ which is important in general, even though these matrices happen to commute and happen to be their own inverses.

We get $Q = (Q_2Q_1)^{-1} = \frac{1}{2} \begin{bmatrix} -1 & -1 & -1 & -1 \\ -1 & -1 & 1 & 1 \\ -1 & 1 & 1 & -1 \\ -1 & 1 & -1 & 1 \end{bmatrix}$ and $R = Q_2Q_1C = \begin{bmatrix} -2 & -4 \\ 0 & -6 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$.

At this point, we check that the diagonal elements of R are non-negative reals. They are not, so negate both Q and R. In general, negate **UPDATE: rows** of R and corresponding columns of Q.

Question 3 (1 points)

Find a 4-term polynomial $p(z)$ through the following points:

z	$p(z)$
1	4
i	8
-1	-4
$-i$	12

UPDATE: Use a form of the FFT, and illustrate how the method is efficient on larger examples.

Solution: Interpolate through ± 1 by $p_0(z) = 4\frac{1+z}{2} + (-4)\frac{1-z}{2} = 4z$ (which clearly checks out). Interpolate $p_1(1) = 8$ and $p_1(-1) = 12$ by $q(z) = 8\frac{1+z}{2} + 12\frac{1-z}{2} = 10 - 2z$, then twiddle to $p_1(z/i) = 10 + 2iz$ to interpolate the given data at $\pm i$. Then combine by

$$\begin{aligned} p(z) &= p_0(z)\frac{1+z^2}{2} + p_1(z/i)\frac{1-z^2}{2} \\ &= (2z + 2z^3) + (5 + iz - 5z^2 - iz^3) \\ &= 5 + (2+i)z - 5z^2 + (2-i)z^3. \end{aligned}$$

Question 4 (1 points)

Recall that, if $Av = \lambda v$, then v is an eigenvector of A with corresponding eigenvalue λ. Find eigenvectors and eigenvalues of

$$\begin{bmatrix} -1 & 1 & & \\ & -1 & 1 & \\ & & -1 & 1 \\ 1 & & & -1 \end{bmatrix}$$

Solution: The above matrix is $S - S^0 = S - I$, where S is the shift operator. The eigenvectors for S are the columns of F_4 (when F_4 is given in doubly-Vandermonde order) and the corresponding eigenvalues are the roots of unity, $1, i, -1, -i$. So the eigenvectors of $S - I = p(S)$ for $p(z) = z - 1$, are the same columns of F_4 with corresponding eigenvalues $1 - 1 = 0, i - 1, -1 - 1 = -2$, and **UPDATE:** $-1 - i$.

For example, $\begin{bmatrix} -1 & 1 & & \\ & -1 & 1 & \\ & & -1 & 1 \\ 1 & & & -1 \end{bmatrix} \begin{bmatrix} 1 \\ i \\ -1 \\ -i \end{bmatrix} = \begin{bmatrix} i-1 \\ -1-i \\ -i+1 \\ 1+i \end{bmatrix} = (i-1) \begin{bmatrix} 1 \\ i \\ -1 \\ -i \end{bmatrix}$.

Question 5 (1 points)

For $A = \begin{bmatrix} 2 & -1 \\ -3 & 5 \\ 0 & 6 \end{bmatrix}$, find $\|A\|_\infty$ and a witnessing $x \neq 0$ that maximizes $\frac{\|Ax\|_\infty}{\|x\|_\infty}$. Briefly explain.

Solution: We want $|-3| + |5| = 8$, witnessed by $x = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, whose signs match those in the witnessing row.

We may divide out by $\|x\|_\infty$, which amounts to maximizing over $\|x\|_\infty = 1$, which means all components in x have magnitude at most 1.

To see this, note that the components of Ax are computed from x and one row at a time of A, so the maximum is the maximum over A-rows r^* of the maximum over x of r^*x. Fix a row r^* and consider the inner maximization. The best x has components ± 1 matching signs in r^*, since that makes all components of r contribute in a reinforcing way rather than canceling. On the other hand, making components of x larger in magnitude than 1 is infeasible and smaller in magnitude than 1 (with the same signs) is suboptimal.

Question 6 (4 points)
Suppose A is given by its SVD [**UPDATE: reduced SVD**] as

$$A = \left(\frac{1}{3}\begin{bmatrix}1 & -2\\ 2 & 1\\ 2 & 0\\ 0 & 2\end{bmatrix}\right)\left(3\sqrt{2}\begin{bmatrix}3 & 0\\ 0 & 2\end{bmatrix}\right)\left(\frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ -1 & 1\end{bmatrix}\right)$$

.

Note the columns of the left parenthesized expression are orthonormal, as are the rows of the right parenthesized expression.

(a) (1 point) Write A as a sum of rank-1 matrices.

Solution: We can read $\sigma_1 u_1 v_1^* + \sigma_2 u_2 v_2^*$. This matrix has the same effect as the given $U\Sigma V^*$ on columns of V (a basis), since $U\Sigma V^* V = U\Sigma$, which scales columns of U; note that $(\sigma_1 u_1 v_1^* + \sigma_2 u_2 v_2^*)v_1 = \sigma_1 u_1$, since $v_1^* v_1 = 1$ and $v_2^* v_1 = 0$. (Take all the columns of V together, in parallel, for $U\Sigma V^*$ and take the columns one at a time for the rank-1 expansion.) We get $3\begin{bmatrix}1\\2\\2\\0\end{bmatrix}\begin{bmatrix}1 & 1\end{bmatrix} + 2\begin{bmatrix}-2\\1\\0\\2\end{bmatrix}\begin{bmatrix}-1 & 1\end{bmatrix}$, canceling the $3\sqrt{2}$ in Σ with $\frac{1}{3}$ in U and $\frac{1}{\sqrt{2}}$ in V^*.

(b) (1 point) Find $\|A\|_2$, using the geometric interpretation of SVD that shows A taking a circle to an ellipse. Briefly explain.

Solution: The 2-norm of a matrix is the largest stretch factor in 2-norm, which is the length of the semi-major axis of the image ellipse. In our geometric view, that's the largest singular value. So $\|A\|_2 = 9\sqrt{2}$.
Alternatively, since U and V^* are unitary, they don't change lengths, so $\|A\|_2 = \|\Sigma\|_2$. If $x^2+y^2 = 1$, then $\Sigma\begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}9\sqrt{2}x\\2y\end{bmatrix}$ and the norm squared is $162x^2 + 4y^2 = 158y^2 + 4(x^2+y^2) = 158x^2 + 4$, which is maximized if x^2 is as large as possible, so $\begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}\pm1\\0\end{bmatrix}$. That picks out $\sigma_1 = 9\sqrt{2}$ from Σ as the vector-2-norm of $\Sigma\begin{bmatrix}\pm1\\0\end{bmatrix}$.

(c) (1 point) What is the rank-1 approximation $\widetilde{A}$ to A that minimizes $\|\widetilde{A} - A\|_2$? For that $\widetilde{A}$, what is the error $\|\widetilde{A} - A\|_2$? Briefly explain, assuming SVDs are unique up to shifting of complex unit factors and the like.

Solution: Set to zero all but the σ_1 term, getting $\sigma_1 u_1 v_1^* = 3\begin{bmatrix}1\\2\\2\\0\end{bmatrix}\begin{bmatrix}1 & 1\end{bmatrix}$. A reduced SVD for $\widetilde{A}$ is $\widetilde{A} = U\left(3\sqrt{2}\begin{bmatrix}3 & 0\\0 & 0\end{bmatrix}\right)V^*$ and a reduced SVD for $A - \widetilde{A}$ is $A - \widetilde{A} = U\left(3\sqrt{2}\begin{bmatrix}0 & 0\\0 & 2\end{bmatrix}\right)V^*$. So $\|A - \widetilde{A}\|_2 = \left\|\begin{bmatrix}0 & 0\\0 & 6\sqrt{2}\end{bmatrix}\right\|_2 = 6\sqrt{2}$, just like the 1-by-1 case. In fact, tracing $A - \widetilde{A}$ on the other column of V, proprotional to $\begin{bmatrix}-1\\1\end{bmatrix}$, we get

$$(A-\widehat{A})\begin{bmatrix}-1\\1\end{bmatrix} = U\left(3\sqrt{2}\begin{bmatrix}0 & 0\\0 & 2\end{bmatrix}\right)V^*\begin{bmatrix}-1\\1\end{bmatrix} = U\left(3\sqrt{2}\begin{bmatrix}0 & 0\\0 & 2\end{bmatrix}\right)\left(\sqrt{2}\begin{bmatrix}0\\1\end{bmatrix}\right) = 6U\begin{bmatrix}0\\2\end{bmatrix},$$

12 times the (unit-length) second column of U, so $A-\widetilde{A}$ takes the vector $\begin{bmatrix}-1\\1\end{bmatrix}$ of length $\sqrt{2}$ to a vector of length 12, which is $\frac{12}{\sqrt{2}} = 6\sqrt{2}$ times bigger.

(d) (1 point) For $b = \begin{bmatrix}8\\-13\\-9\\-8\end{bmatrix}$, find the x that minimizes $\|Ax-b\|_2$, illustrating how to use the SVD to solve faster than from scratch, in general.

Solution: Since the range of A and the range of U are the same, we can take UU^* as the projection onto the range of A. The best value of Ax is UU^*b.
As the problem is written, A is actually given by a **Reduced** SVD $A = \widehat{U}\widehat{\Sigma}V^*$. The matrix $\widehat{U}$ is oblong, so not invertible, but $\widehat{U}^*\widehat{U} = I_2$, and so we can peel off $\widehat{U}$ anyway. So $\widehat{U}\widehat{U}^*b$ (which is the projection of B even for oblong $\widehat{U}$) is to be Ax, so $\widehat{U}\Sigma V^*x = \widehat{U}\widehat{U}^*b$. Multiplying by $\widehat{U}^*$ on the left, we get $\Sigma V^*x = \widehat{U}^*b$. We can invert Σ and V to isolate $x = V\Sigma^{-1}U^*b$.
Note that inverting diagonal Σ involves reciprocating the diagonal elements and the inverse of V^* is the conjugate-transpose. Similarly, $\widehat{U}$ is canceled (not quite inverted) by $\widehat{U}^*$. All these operations take time at most the number of components of A, which is mn, and less than the cost of solving $Ax = b$ for square A (a special, easier case of least squares).

Moving constants around, $A = \begin{bmatrix}1 & -2\\2 & 1\\2 & 0\\0 & 2\end{bmatrix}\begin{bmatrix}3 & 0\\0 & 2\end{bmatrix}\begin{bmatrix}1 & 1\\-1 & 1\end{bmatrix}$.

We get: $x = \frac{1}{18}\begin{bmatrix}1 & -1\\1 & 1\end{bmatrix}\begin{bmatrix}1/3 & 0\\0 & 1/2\end{bmatrix}\begin{bmatrix}1 & 2 & 2 & 0\\-2 & 1 & 0 & 2\end{bmatrix}\begin{bmatrix}8\\-13\\-9\\-8\end{bmatrix}$.

To evaluate, do the rightmost multiplication first, to move away from dimension 4, so

$$x = \frac{1}{18}\begin{bmatrix}1 & -1\\1 & 1\end{bmatrix}\begin{bmatrix}1/3 & 0\\0 & 1/2\end{bmatrix}\begin{bmatrix}-36\\-45\end{bmatrix} = \frac{1}{36}\begin{bmatrix}1 & -1\\1 & 1\end{bmatrix}\begin{bmatrix}-24\\-45\end{bmatrix} = \frac{1}{36}\begin{bmatrix}21\\-69\end{bmatrix} = \frac{1}{12}\begin{bmatrix}7\\-23\end{bmatrix}.$$

This gives $Ax = \begin{bmatrix}6\\-13\\-8\\-10\end{bmatrix} \approx \begin{bmatrix}8\\-13\\-9\\-8\end{bmatrix}$, which is reasonable. The residual is $\begin{bmatrix}8\\-13\\-9\\-8\end{bmatrix} - \begin{bmatrix}6\\-13\\-8\\-10\end{bmatrix} = \begin{bmatrix}2\\0\\-1\\2\end{bmatrix}$, which is indeed orthogonal to both columns of A.